Class Manual 3

1. **Display all students & courses with enrollment status**

   QUERY

```
SELECT S.name AS student_name,
       C.course_name,
       CASE
           WHEN E.student_id IS NULL THEN 'Not Enrolled'
           ELSE 'Enrolled'
       END AS status
FROM STUDENT S
CROSS JOIN COURSE C
LEFT JOIN ENROLLMENT E
ON S.student_id = E.student_id
AND C.course_id = E.course_id;
```

   OUTPUT

	student_name	course_name	status
▶	Aman	Data Science	Not Enrolled
	Aman	AI	Not Enrolled
	Aman	Networks	Not Enrolled
	Aman	OS	Enrolled
	Aman	DBMS	Enrolled
	Riya	Data Science	Not Enrolled
	Riya	AI	Not Enrolled
	Riya	Networks	Not Enrolled
	Riya	OS	Enrolled
	Riya	DBMS	Enrolled
	Karan	Data Science	Not Enrolled
	Karan	AI	Not Enrolled
	Karan	Networks	Not Enrolled
	Karan	OS	Enrolled
	Karan	DBMS	Enrolled
	Neha	Data Science	Enrolled
	Neha	AI	Not Enrolled
	Neha	Networks	Not Enrolled
	Neha	OS	Not Enrolled
	Neha	DBMS	Enrolled
	Rahul	Data Science	Not Enrolled
	Rahul	AI	Not Enrolled
	Rahul	Networks	Enrolled
	Rahul	OS	Not Enrolled
	Rahul	DBMS	Not Enrolled
	Sneha	Data Science	Not Enrolled
	Sneha	AI	Not Enrolled
	Sneha	Networks	Enrolled
	Sneha	OS	Not Enrolled
	Sneha	DBMS	Enrolled
	Arjun	Data Science	Not Enrolled
	Arjun	AI	Enrolled
	Arjun	Networks	Not Enrolled
	Arjun	OS	Not Enrolled
	Arjun	DBMS	Enrolled
	Priya	Data Science	Enrolled
	Priya	AI	Not Enrolled
	Priya	Networks	Not Enrolled
	Priya	OS	Not Enrolled
	Priya	DBMS	Not Enrolled

2. **Find courses taught by professors with more than 1 section**

QUERY

```
SELECT P.name AS professor_name,
       C.course_name,
       COUNT(T.section) AS total_sections
FROM TEACHES T
JOIN PROFESSOR P ON T.professor_id = P.professor_id
JOIN COURSE C ON T.course_id = C.course_id
GROUP BY P.name, C.course_name
HAVING COUNT(T.section) > 1;
```

OUTPUT

	professor_name	course_name	total_sections
▶	Dr. Sharma	DBMS	2
	Dr. Mehta	OS	2

3. **Count all possible student course pairs**

QUERY

```
SELECT COUNT(*) AS total_pairs
FROM STUDENT S
CROSS JOIN COURSE C;
```

OUTPUT

	total_pairs
▶	40